

Eyal Itskovits

Machine Learning Scientist specializing in Drug Discovery & Computational Biology

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PROFESSIONAL EXPERIENCE

2024-PRESENT

Director of Machine Learning and Artificial Intelligence

Leading GSK Flagship AI Scientist project

Leading a multi-team, globally distributed project, with direct reporting and engagement with GSK's Global Leadership Team (including CEO) and GSK's Board. The project aims to design a generative AI-based hypothesis generation engine that leverages internal and external data sources to enable scaled, efficient scientific discovery.

2023-2024

Senior AIML Research Engineer and Team Leader

Team leading, AIML Research

Led a team of six PhD-level researchers building ML models for biological perturbation prediction and biomarker discovery in oncology. Supervised and mentored UK-based PhD student interns.

2021-2023

Artificial Intelligence and Machine Learning Research Engineer

Led the development of advanced ML models for predicting cancer-type-specific responses to genetic perturbations. Combined cutting-edge machine learning research with deep collaboration across oncology teams and experimental labs.

EDUCATION

PhD, Computer Science and Computational Biology, 2021

Hebrew University of Jerusalem

Thesis: "Sensation of complex stimuli in the *C. elegans* nematode"

MSc, Computer Science and Computational Biology, 2015

Hebrew University of Jerusalem

Research topic:

- Prof. Alon Zaslaver - Developing machine learning models for tracking and characterizing high-throughput behavioral experiments

Rotational research:

- Prof. Sagiv Shifman - Modeling complex genetic effects underlying Autism Spectrum Disorder (ASD)
- Prof. Nir Friedman - Modeling epigenetic effects on gene expression

BSc, Computer Science and Computational Biology, 2013

Hebrew University of Jerusalem

PROFESSIONAL SERVICE

Conference Reviewer:

- Neural Information Processing Systems (NeurIPS), 2024
- International Conference on Learning Representations (ICLR) - ML For Drug Discovery Workshop, 2023

TEACHING EXPERIENCE

Probability and Statistics, 2014-2019

Hebrew University - Course 72301

Object Oriented Programming, 2013

Hebrew University - Course 67125

General Genetics, 2012

Hebrew University - Course 72155

SKILLS

Machine Learning Research and Engineering

Extensive practical experience and a strong theoretical understanding of:

- **Probabilistic Deep Learning**
- **Generative Deep Learning**
- **Signal processing** and **time-series** analysis
- **Graph Neural Networks**
- Pre-Training and fine-tuning of **large foundation models**

Software Development

- Expert in **Python** for scientific computing and ML research; experience with **C/C++** for performance-critical applications
- Proficient with modern ML frameworks and scientific computing tools
- Experience with **operating systems** internals. Strong expertise with **Linux** environments and **cloud computing** platforms
- Experience with **application development lifecycle management**, including version control, testing, and deployment pipelines

Life Sciences

- Deep understanding of **drug discovery pipelines**, including target identification, validation, and biomarker discovery in **oncology**
- Expertise in **cancer biology**, genetic perturbations, and cellular response mechanisms
- Analysis and integration of **multi-omics data** (genomics, transcriptomics, proteomics)
- Experience with high-dimensional **neural activity data** analysis
- Experience with advanced **optical systems** and **microscopy**

HONORS AND AWARDS

- ▶ JBC Postdoctoral Fellowship - 2020
- ▶ HHMI Janelia's Junior Scientist Workshop on Machine Learning and Computer Vision participant - 2019
- ▶ 'Next Generation of Neuroscience' workshop for Exceptional PhD Students - 2019
- ▶ The Irene and Michael Mochrik Award for Outstanding Genetics Students - 2018
- ▶ SMART Award, Alexander Silverman Institute of Life Sciences - 2018
- ▶ Excellent Teaching Assistant - 2013-2016, 2018-2019
- ▶ JBC Golden PhD Scholarship - 2017
- ▶ Best Poster - I-Core Single Molecule Biophysics Conference - 2015
- ▶ JBC Travel Scholarship - 2015
- ▶ Honorable Mention - International *C.elegans* Meeting - 2015
- ▶ Mada - Life Sciences Excellence Program - 2013
- ▶ Dean's List - 2010-2012, 2013-2015

VOLUNTEERING

2016-2017 "Aleh" Running Team - Serving as a guide runner for blind and visually impaired athletes

2015-2017 "Magdilim Rosh" - Exposing high school students to neuroscience research

2014 "Exposure to Science" - Introducing research and basic science topics to adolescents from complex backgrounds

2011 "Magshimim Program" - Teaching computer science and information security to high school students from the social periphery

ONLINE PROFILES

Google Scholar

LinkedIn

GitHub

PUBLICATIONS

- **E Itskovits**, R Ruach, A Kazakov, A Zaslaver. 2018.
"Concerted pulsatile and graded neural dynamics enables efficient chemotaxis in *C. elegans*."
Nature Communications 9 (1), 2866.
- **E Itskovits**, A Levine, E Cohen, A Zaslaver. 2017.
"A multi-animal tracker for studying complex behaviors."
BMC Biology 15 (1), 29.
- A Azulay, **E Itskovits**, A Zaslaver. 2016.
"The *C. elegans* Connectome Consists of Homogenous Circuits with Defined Functional Roles."
PLoS Computational Biology 12 (9), e1005021.
- D Mukherjee, BM Ignatowska-Jankowska, **E Itskovits**, BJ Gonzales, et al. 2018.
"Salient experiences are represented by unique transcriptional signatures in the mouse brain."
eLife 7, e31220.
- S Iwanir, R Ruach, **E Itskovits**, CO Pritz, E Bokman, A Zaslaver. 2019.
"Irrational behavior in *C. elegans* arises from asymmetric modulatory effects within single sensory neurons."
Nature Communications 10 (1), 3202.
- C Pritz, **E Itskovits**, E Bokman, R Ruach, V Gritsenko, T Nelken, et al. 2023.
"Principles for coding associative memories in a compact neural network."
eLife 12, e74434.
- N Deshe, Y Eliezer, L Hoch, **E Itskovits**, E Bokman, S Ben-Ezra, et al. 2023.
"Inheritance of associative memories and acquired cellular changes in *C. elegans*."
Nature Communications 14 (1), 4232.
- H Ishtayeh, H Achache, E Kroizer, Y Rappaport, **E Itskovits**, H Gingold, et al. 2021.
"Systematic analysis of long intergenic non-coding RNAs in *C. elegans* germline uncovers roles in somatic growth."
RNA Biology 18 (3), 435-445.
- E Bokman, CO Pritz, R Ruach, **E Itskovits**, H Sharvit, A Zaslaver. 2024.
"Intricate response dynamics enhances stimulus discrimination in the resource-limited *C. elegans* chemosensory system."
BMC Biology 22 (1), 173.
- R Ruach, S Yellinek, **E Itskovits**, N Deshe, Y Eliezer, E Bokman, et al. 2022.
"A negative feedback loop in the GPCR pathway underlies efficient coding of external stimuli."
Molecular Systems Biology 18 (9), e10514.
- A Bean, A Weiner, A Hughes, **E Itskovits**, N Friedman, OJ Rando. 2016.
"Genome-wide histone modification patterns in *Kluyveromyces Lactis* reveal evolutionary adaptation of a heterochromatin-associated mark."
bioRxiv, 039776.
- I Kirshenboim, B Aviner, **E Itskovits**, A Zaslaver, L Broday. 2021.
"Dopamine-dependent biphasic behaviour under 'deep diving' conditions in *Caenorhabditis elegans*."
Proceedings of the Royal Society B 288 (1946), 20210128.